St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Musculoskeletal System | Lecture 03

Clinical Anatomy of the Back

Dr. Mohamed Abdelrahim

mabdelra@sgu.edu

Department of Anatomical Sciences

School of Medicine, St. George's University

Copyright

Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 5th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 1: pg 7- 13
- Online Textbook: Chapter 1 https://www.clinicalkey.com/student/content/book/3-s2.0-B9780323934237000022

- Pay special attention to the "**In the Clinic**" boxes for the clinical correlations for this topic
  - Blue boxes in online text
  - Green boxes in hard copy

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1108	Describe the anatomical location and mechanism of hyperextension injuries of the neck.
SOM.MKII.BPM1.1.MSK.1.ANAT.1121	Describe and identify (in imaging studies) the anatomical location, mechanism and clinical sequences of a Jefferson fracture, compression fractures of the vertebrae including wedge and Hangman's fracture
SOM.MKII.BPM1.1.MSK.1.ANAT.1114	Describe the anatomical location, mechanism and clinical sequences of fractures of the dens and identify them in imaging studies
SOM.MKII.BPM1.1.MSK.1.ANAT.1109	Describe and identify brevicollis (Klippel-Feil) and hemivertebra and explain how they can lead to scoliosis (Grey's Anatomy For Students "In the Clinic" Blue Boxes)
SOM.MKII.BPM1.1.MSK.1.ANAT.1110	Describe sacralization of the lumbar vertebrae and lumbarization of the sacral vertebrae (Grey's Anatomy For Students "In the Clinic" Blue Boxes)
SOM.MKII.BPM1.1.MSK.1.ANAT.1115	Describe the process of disc herniation and its consequences in terms of nerve impingements paying particular attention to the differences between the cervical, thoracic and lumbar regions.
SOM.MK..BPM1.1.MSK.1.ANAT.1116	Identify herniation using radiographic studies and use this information to interpret clinical signs and symptoms
SOM.MKII.BPM1.1.MSK.1.ANAT.1117	Describe the anatomical sequelae after rupturing the transverse ligament of the atlas.
SOM.MKII.BPM1.1.MSK.1.ANAT.1118	Describe the anatomical landmarks used for lumbar epidural and sacral (caudal) epidural anesthesia and explain when each will be performed.
SOM.MKII.BPM1.1.MSK.1.ANAT.1120	Describe the mechanism(s) that produces lumbar spinal stenosis and use this information to interpret clinical signs and symptoms (SG Article)
SOM.MKII.BPM1.1.MSK.1.ANAT.1119	List the neurological signs and symptoms of spondylolysis and spondylolisthesis and be able to identify them in imaging studies.

Lecture Overview

- Clinical Case 1
  - Hyperextension Injury of the Neck
  - Transverse Ligament Rupture
  - Dens Fractures
- Clinical Case 2
  - Jefferson Fracture
- Clinical Case 3
  - Disc Herniations
  - Wedge Fracture
- Clinical Case 4
  - Hangman's Fracture
  - Lumbar Vertebrae Burst Fracture
  - Spondylolysis and Spondylolisthesis
- Clinical Case 5
  - Analgesia and Epidural Anesthesia
  - Abnormal Vertebrae Count
  - Klippel-Feil Syndrome
  - Hemivertebrae

Clinical Case 1

A 34-year-old man is brought to the emergency department after being involved in a motor vehicular accident. The patient was a restrained (wearing seat belt) driver of a **rear-ended accident** on the highway. He was found in his car conscious complaining of neck pain. The cervical spine was immobilized. Vital signs on the scene and during transport to hospital were within normal limits.

Reassessment of vital signs shows no significant changes. Physical examination shows **diffuse posterior neck tenderness, limited neck mobility due to pain** and **full range mobility of the upper and lower extremities**.

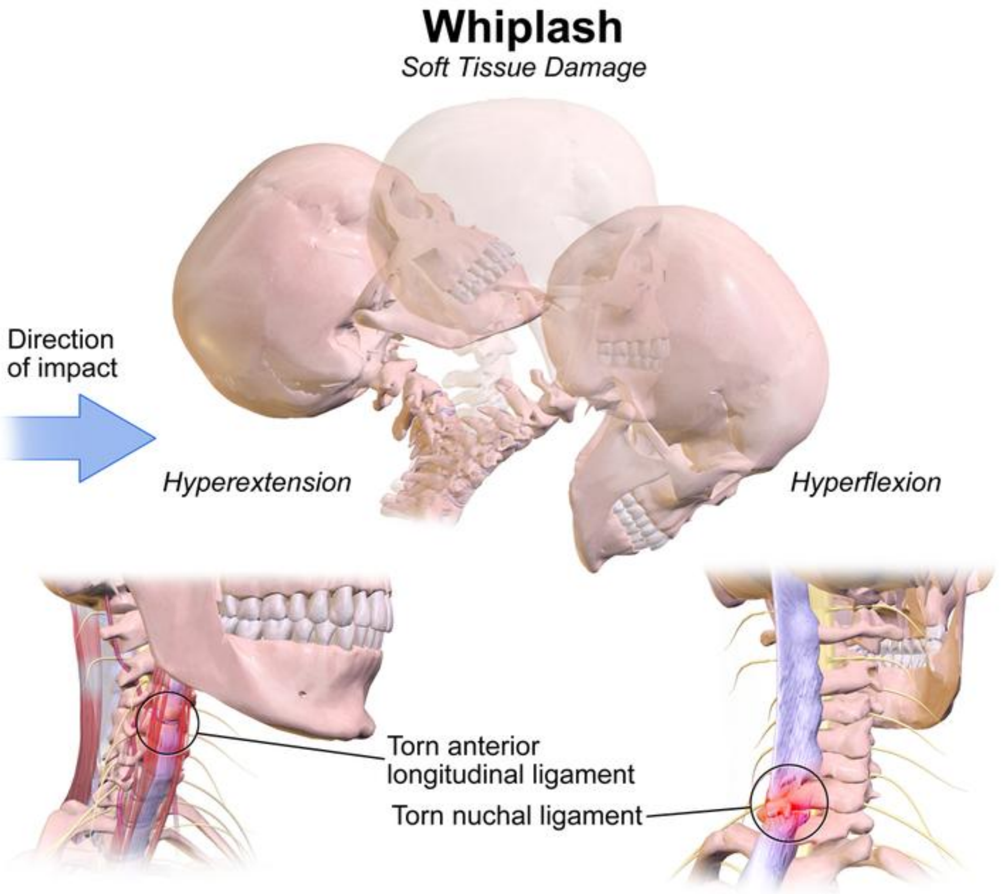
Clinical imaging shows ligamentous strain. He is also **placed on neurological observation.**

Hyperextension Injury of the Neck

- **Contact or Indirect applied force**
  - **Contact →** direct anterior force to the forehead
  - **Noncontact →** unrestrained neck motion (Rear-end motor collision with person wearing seat belt); ***The Whiplash Injury***

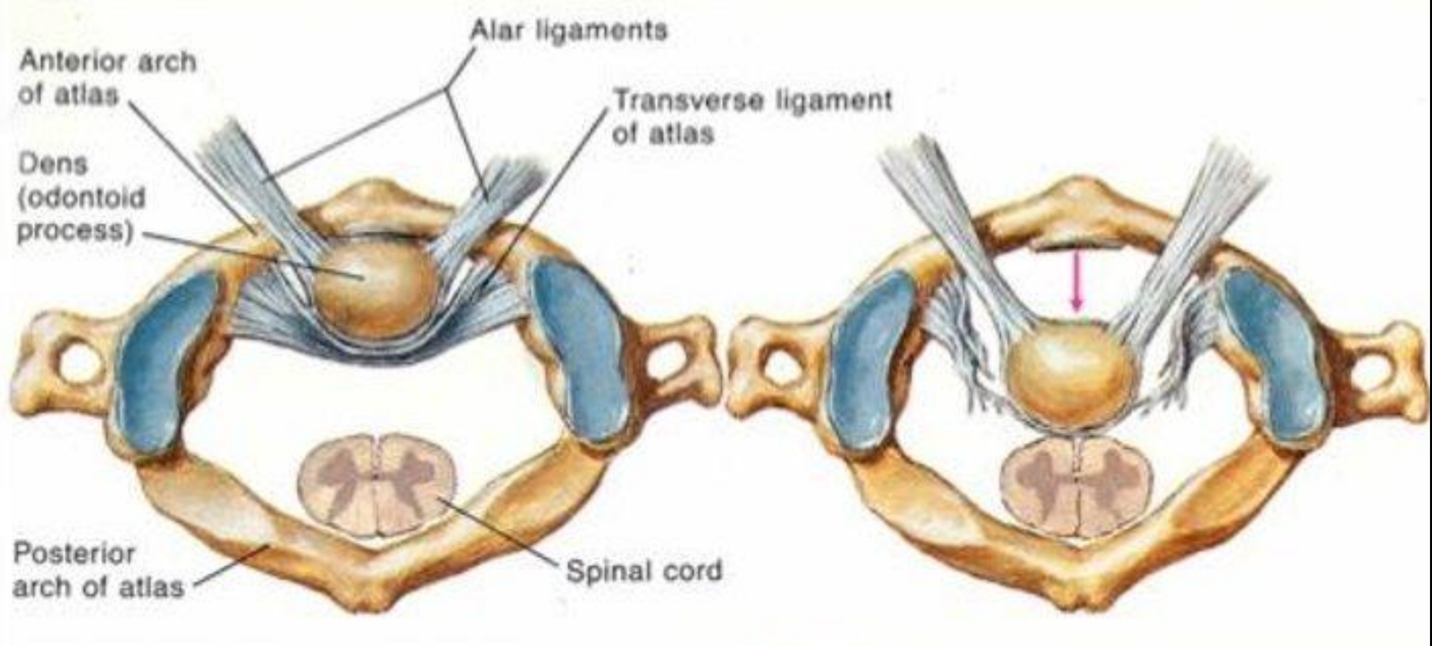
- Typically, ligamentous damage involving the anterior longitudinal ligament (hyperextension) and nuchal ligament (Hyperflexion)

- Headache, **neck pain radiating into shoulders**, upper back pain, dizziness, and neurological signs of a concussion



Transverse Ligament Rupture

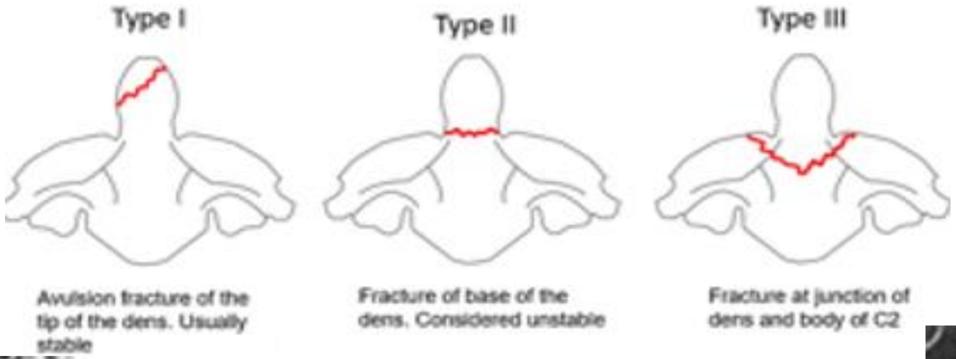
- Can result from whiplash injury
- Ligamentous rupture causes an increase flexion angle of skull and the cervical vertebrae
- Dislocation of the dense into vertebral canal→ Compression of spinal cord
    - Unconscious patient
    - Suppression of diaphragm
    - Paraplegia
    - Incontinence



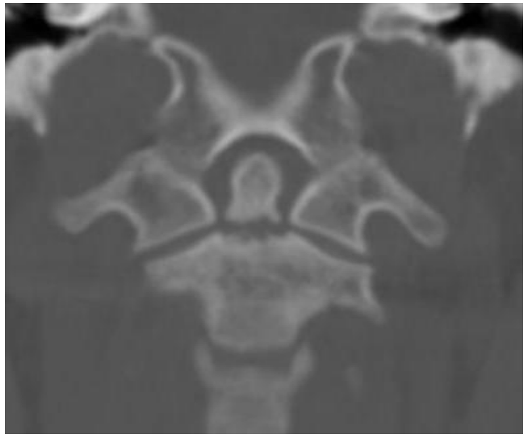
Axial CT Scan, Dens subluxation due to transverse ligament rupture.

Dens Fracture

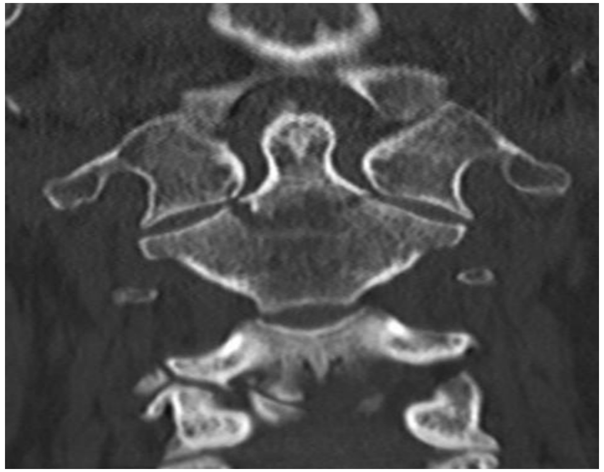
- Dens fracture can occur with flexion or extension injuries
- Avulsion of dens posteriorly can compress the spinal cord → + neurological symptoms
  - Neck pain
  - Numbness in limbs
  - Quadriplegia
  - Respiratory arrest → diaphragm paralysis



Type 1

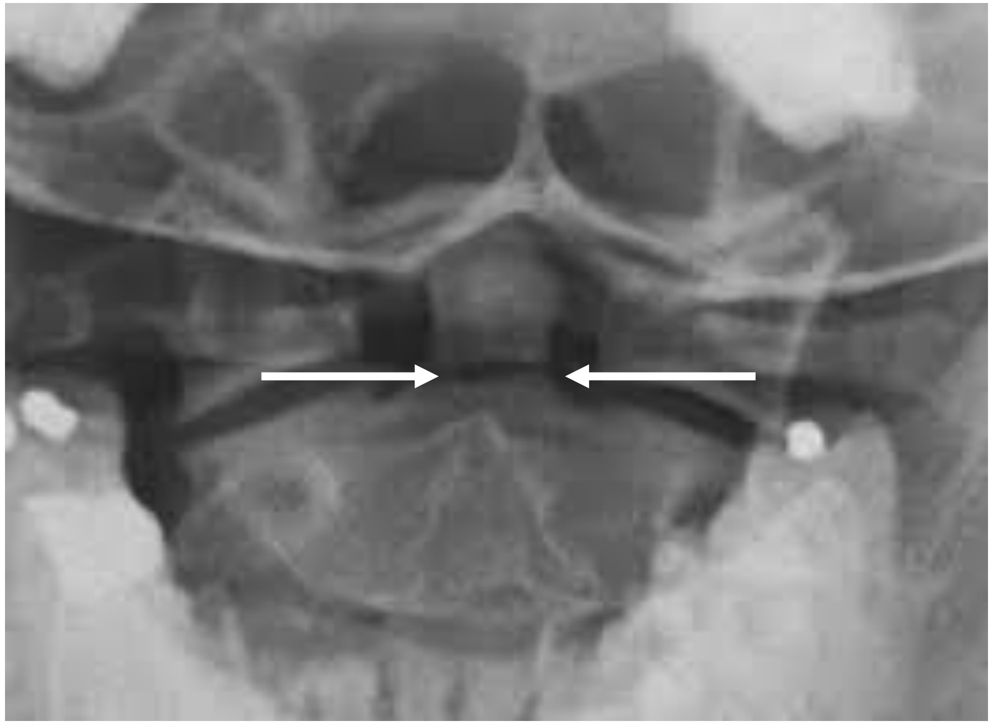


Type 2



Type 3

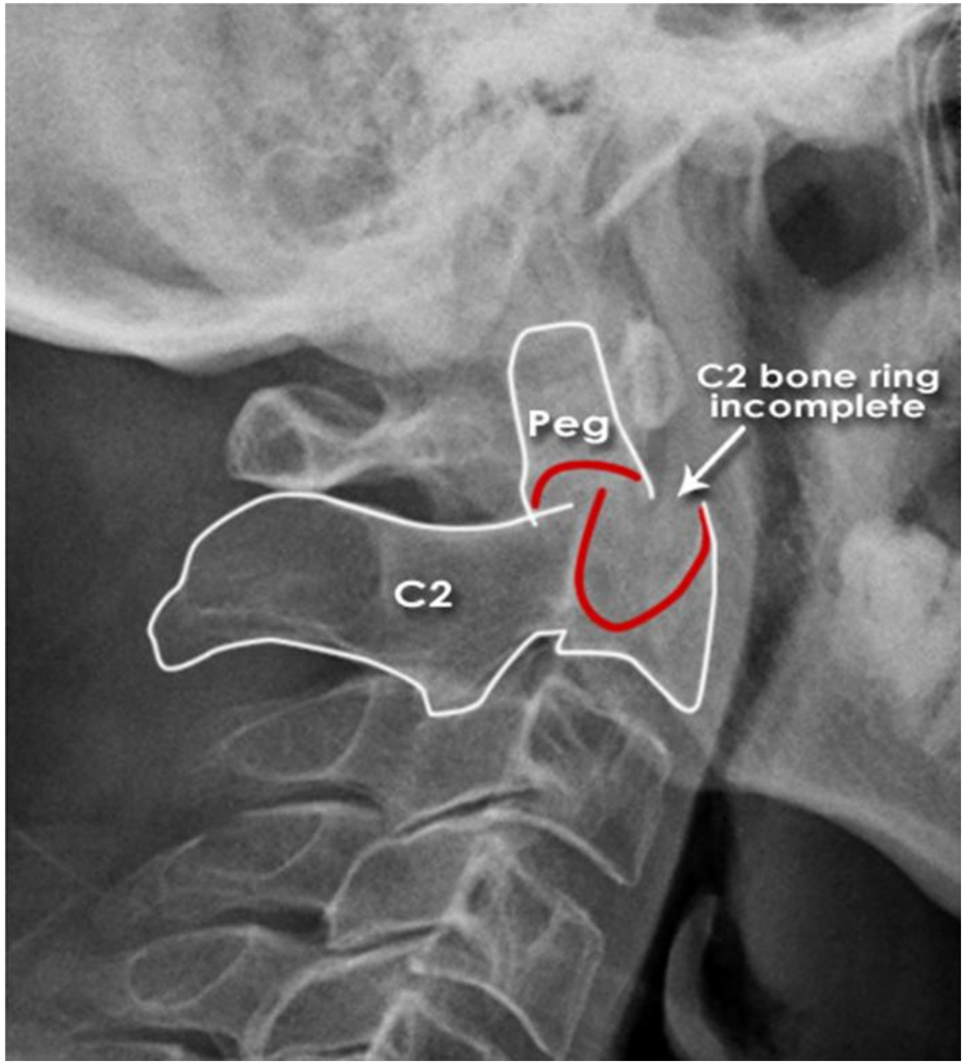
Dens Fracture (Open Mouth View X Ray)



Open mouth view X-ray, fracture of dens. Fracture line depicted between the white arrows

St. George's University
Grenada, West Indies

Dens Fracture (Lateral View X Ray)

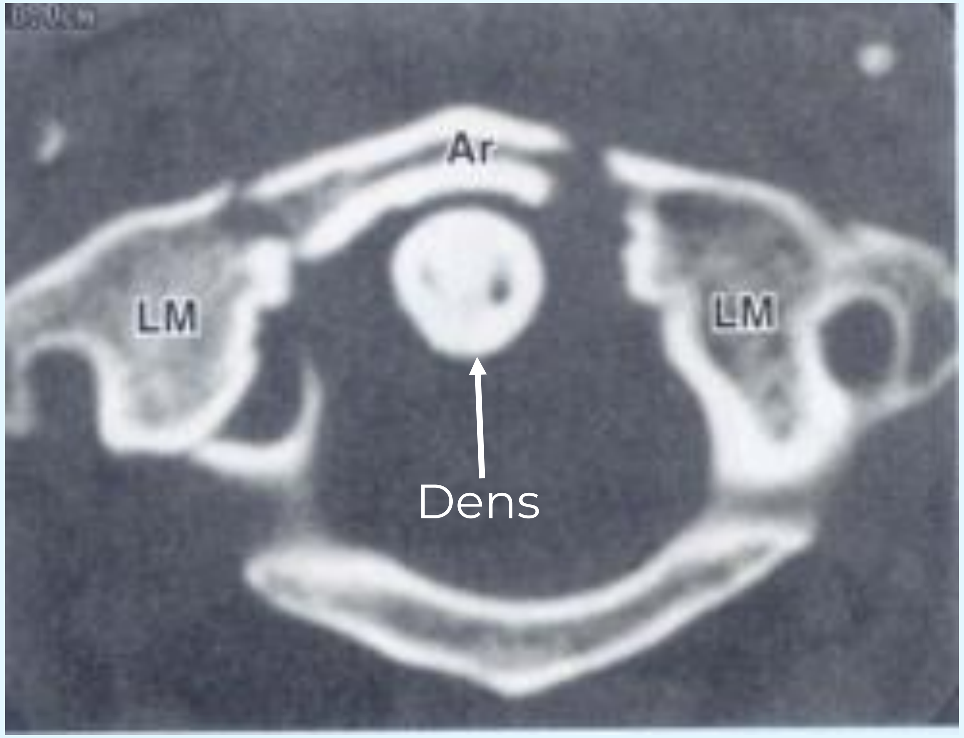


Lateral view X-ray craniocervical junction region, fracture of dens (Type 2)

Clinical Case 2

A 65-old-man is brought to the emergency department 5 days after **falling headfirst** from slipping down a staircase. The day after falling he noticed **difficulty in lifting his head from his chest and neck pain**. The patient proceeded to seek help from a chiropractor who performed **cervical manipulations** after interpreting the patient's X-Rays as normal. However, the patient's symptoms worsen and was referred to a physician.

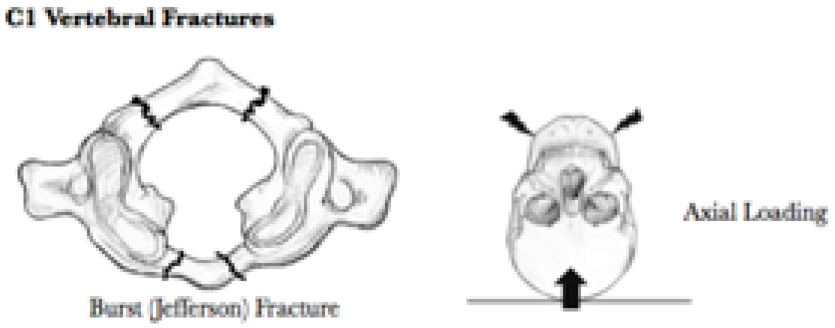
In the emergency room, the patient's vital signs are with in normal limits. Physical examination shows a **cervical tenderness, hesitation in neck flexion and extension; no signs of focal neurological deficits**. A CT scan of the patient is shown below:-



Modified from Haus, B. M., & Harris, M. B. (2008). Case report: nonoperative treatment of an unstable Jefferson fracture using a cervical collar. *Clinical orthopaedics and related research*, *466*(5), 1257–1261. https://doi.org/10.1007/s11999-008-0143-5

Jefferson Fracture

- An example of a **burst fracture; fracture of the C1 Vertebrae**
- **Axial loading injury** → Axial force is transmitted across the occipital-cervical junction (**thinking diving into a shallow pool**)
    - "Atlas gets crushed between skill and Axis" → **break in arches→ lateral masses move outward → look out for transverse ligament injury!**
- Rupture of the transverse ligament is also seen→ highly unstable fracture



"Diving into a shallow pool"

Frontal X-ray Open Mouth View: White arrows showing nonalignment of C1 Lateral masses with C2 indicating loss of integrity of the C1 Boney ring

Image from Haus, B. M., & Harris, M. B. (2008). Case report: nonoperative treatment of an unstable Jefferson fracture using a cervical collar. *Clinical orthopaedics and related research*, *466*(5), 1257–1261. https://doi.org/10.1007/s11999-008-0143-5

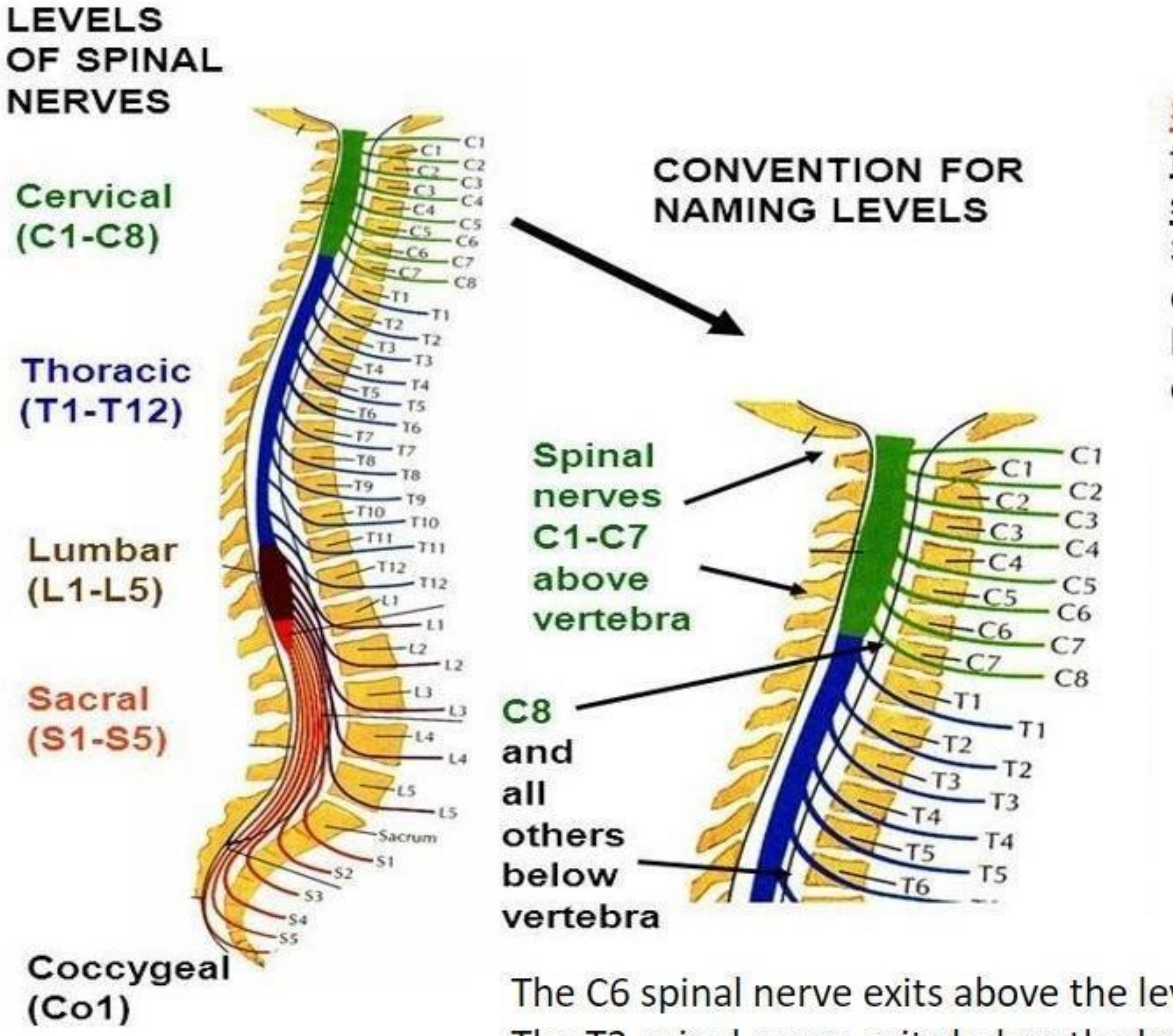
Clinical Case 3

A **70-year-old woman** is brought to the emergency room because of **back pain**. The pain started 6 days after **falling from a chair on her buttocks**. She mentions that **this pain feels different from the usual back pain that shoots down the back of the left leg**.

Physical examination shows a **hunched over posture**; **tenderness over the L4 vertebra; difficulty with lateral flexion, flexion and extension of the lumbar region**.

Clinical imaging of the lumbar spine is ordered to further assess the patient.

Numbering of Spinal Nerves



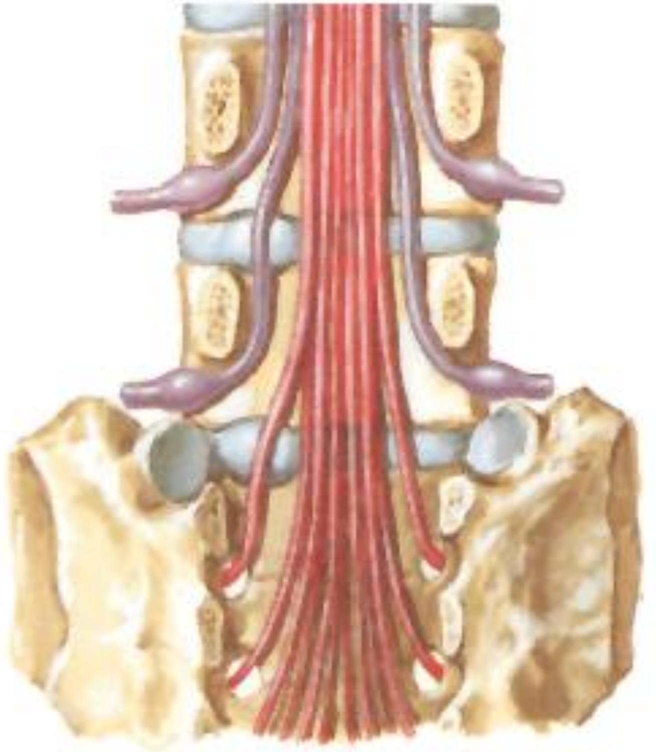
Spinal nerves - arise **from/project to spinal cord**; there are 31 spinal nerves (8 cervical, 12 thoracic, 5 lumbar, 5 sacral and 1 coccygeal).

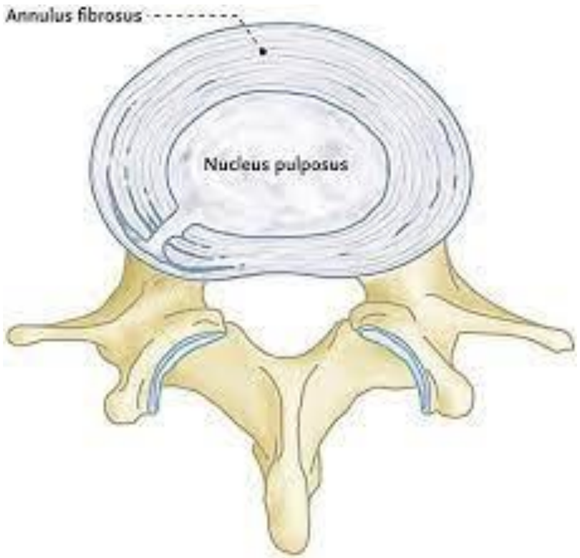
The C6 spinal nerve exits above the level of the C6 vertebra

The T3 spinal nerve exits below the level of the T3 vertebra

Disc Herniations (Lumbar)

- Most commonly occurs in the lumbar region → Intervertebral disc at **L4/L5** and **L5/S1** vertebral levels most commonly affected
  - **L4/L5 disc herniation** pinches the **L5 spinal nerve**
  - **L5/S1 disc herniation** pinches the **S1 spinal nerve**
  - Pain radiates along the dermatomes of the pinched nerve

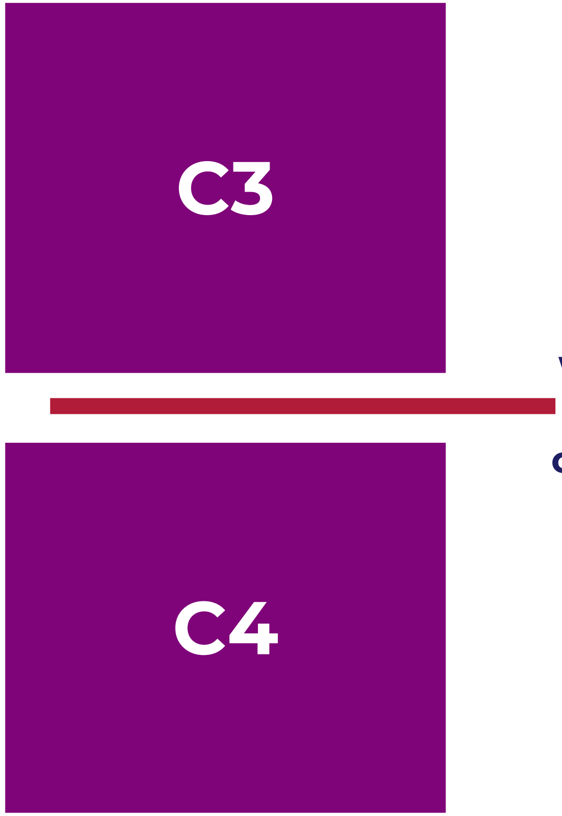




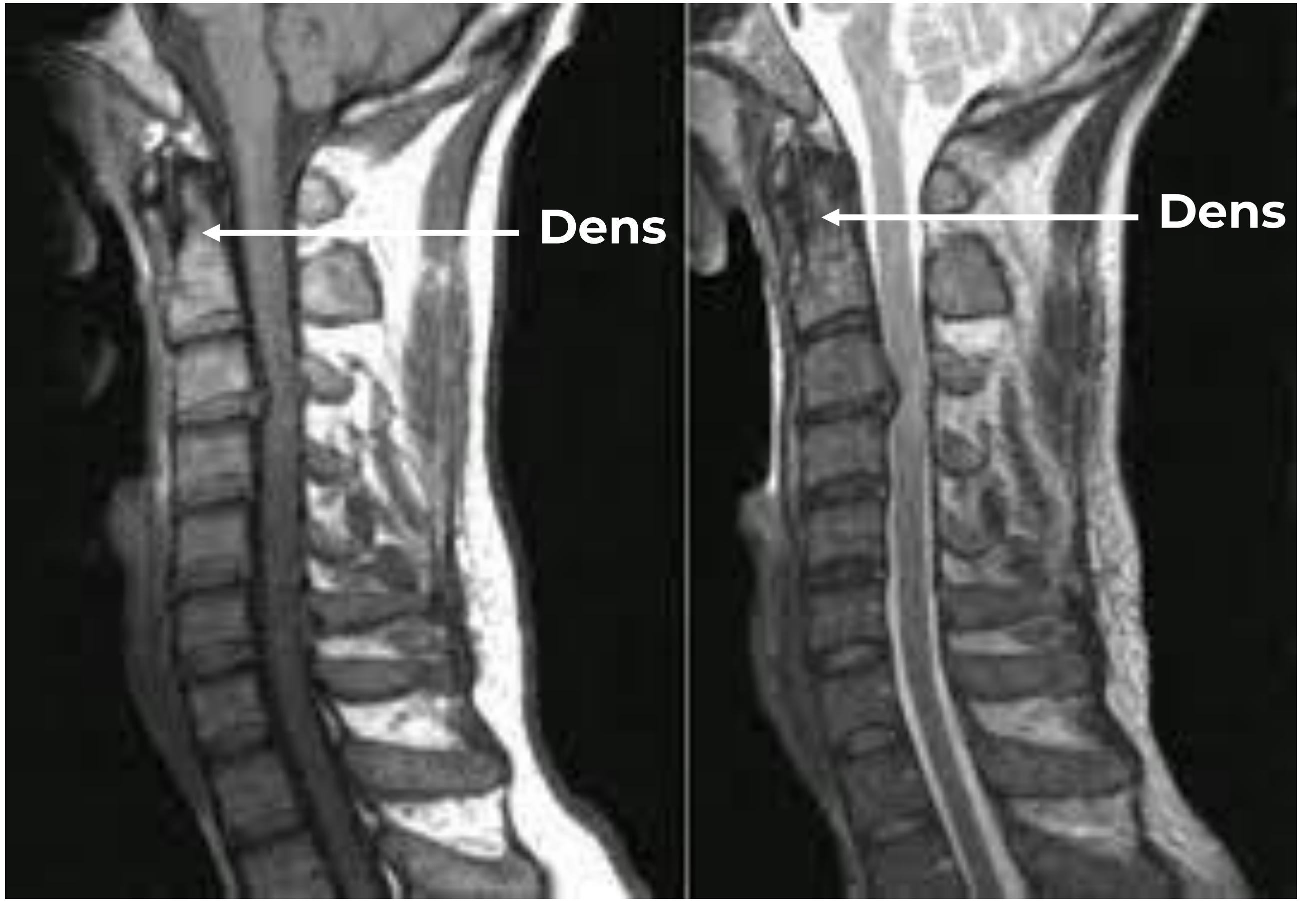
MRI (T2) lumbar disc herniation (L5/S1 disc)

Disc Herniations (Cervical/Thoracic)

- Herniation can also occur in the cervical and thoracic region
  - Spinal nerve **immediately** exiting out of the IV foramen is pinched



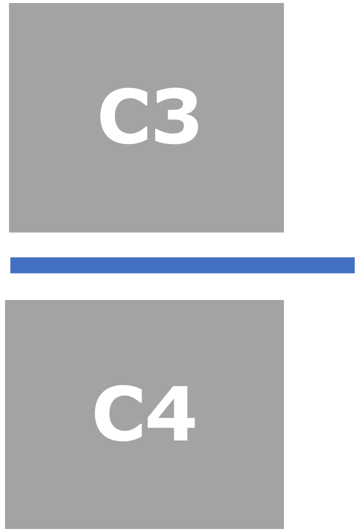
Which nerve is compressed?



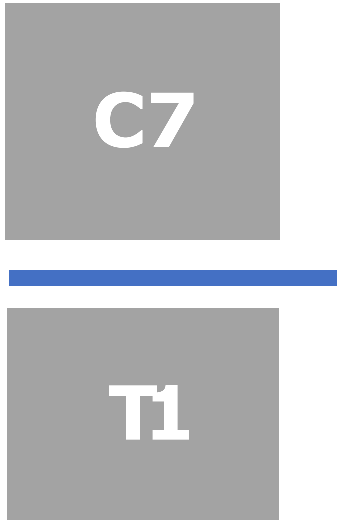
MRI (T1) cervical disc herniation

MRI (T2) cervical disc herniation

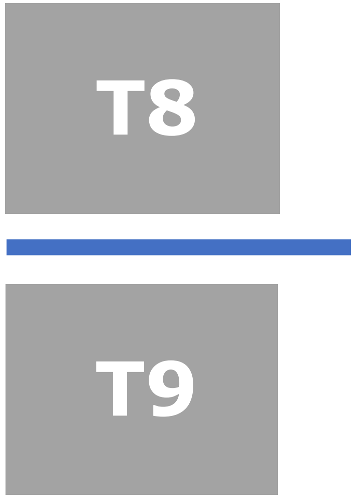
Homework: Using the following illustrations indicate which spinal nerve will most likely be compressed if a herniation is present:



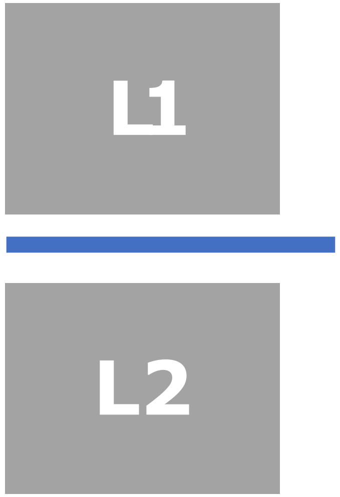
Which nerve is compressed?



Which nerve is compressed?



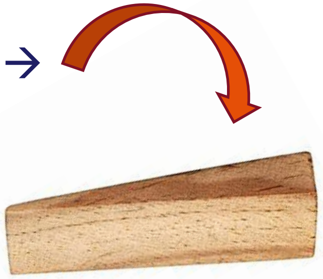
Which nerve is compressed?



Which nerve is compressed?

Wedge Fracture

- Compression fracture of the vertebral body
- Loss of an equal anterior to posterior height of the vertebral body
    - Anterior column fracture but middle column intact → Wedge shaped vertebral body
- Results from an axial force with a compressive load (**hyperflexion**)
- **Osteoporosis** or bone abnormalities (such as cancer metastasis) that weakens bone.
    - Some cases to result from hyperflexion injuries with no underlying bone abnormalities → severe forward bending injury→ Front impact car accidents
- Clinical signs
    - Pain and loss of mobility
    - Kyphosis
    - Loss of height



X-ray lateral view, wedge fracture of L1 vertebrae

Clinical Case 4

Two teenagers (Patients A and B) are brought to the emergency department following a motor vehicular accident.

Patient A was an **unrestrained driver** who was found **unresponsive** with their **neck hyperextended and face smashed into the front windshield, chest on steering wheel**. Cervical spine was immobilized, and the patient was removed from the motor vehicle. After several efforts of resuscitation, the patient **regains consciousness** but was unable to move the upper and lower extremities.

Patient B was a motorcycle rider (wearing helmet) who was **pitched into the air** off the bike. The patient was **found conscious** and vital signs show elevated blood pressure and tachycardia. They complain of **lower back pain and difficulty moving the lower extremity.**

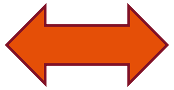
After further assessment, clinical Imaging is ordered to further access the extent of the patients' injuries.

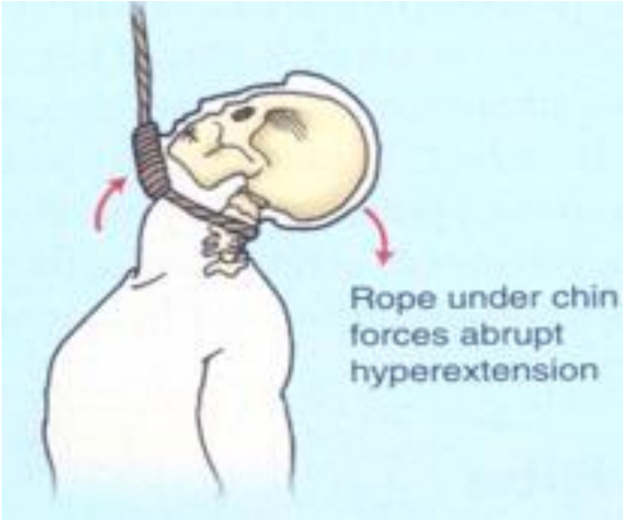
St. George's University
Grenada, West Indies

Hangman's Fracture

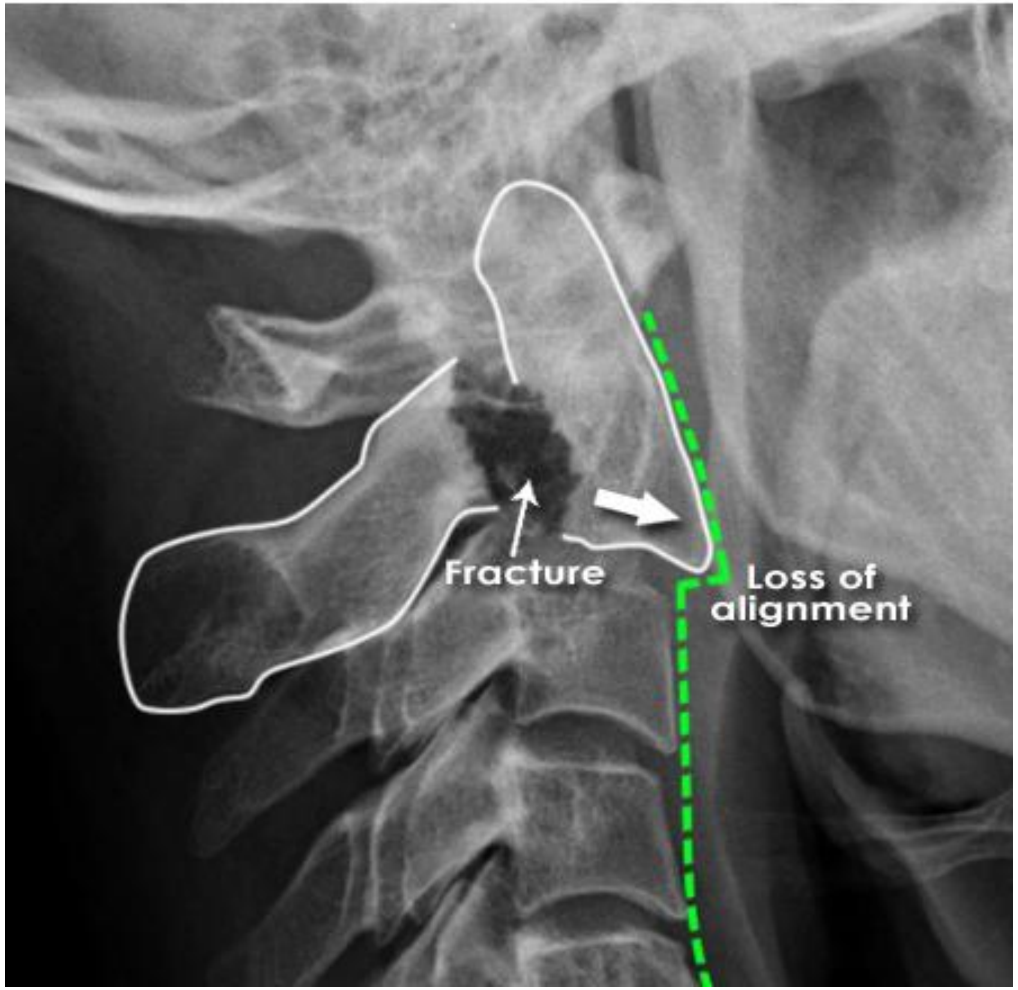
Patient A

- C2 Vertebral Fracture
  - **Bilateral traumatic fracture** of pars interarticularis a/w traumatic subluxation of C2 on C3
- Occurs after forced hyperextension with distraction of neck. Causes fracture because the pedicle
- Motor vehicle accidents (**unrestrained** driver hitting head on dashboard/windshield), diving injuries, or contact sports injuries





Hangman's Fracture



Lateral view of craniocervical junction with pars interarticularis fracture of C2

St. George's University
Grenada, West Indies

Lumbar Vertebrae Burst Fracture

- Compression fracture; Bone is crushed in all directions
- High-energy traumatic vertebral fractures caused by flexion of the spine leading to a compression force through the anterior and middle column
- Reduction of entire vertebra's height
- Retropulsion of bone into vertebral canal → compression of neural structures (spinal cord, spinal nerves and/or roots)

Patient B

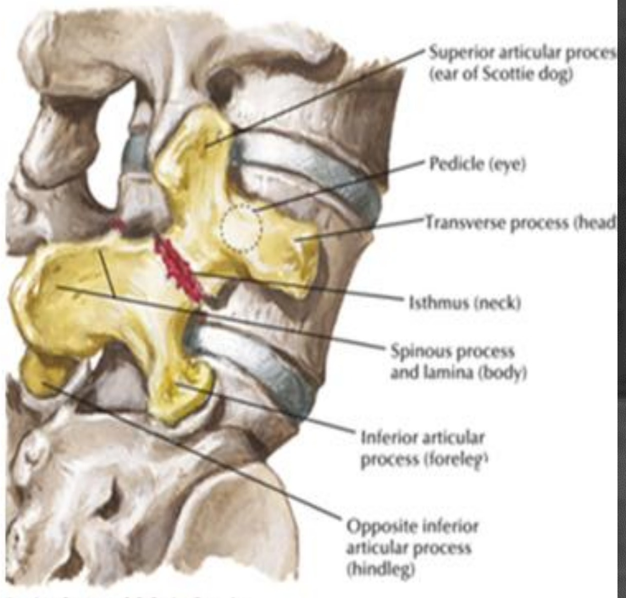
CT Scan lateral view, burst fracture of L1 vertebrae

St. George's University
Grenada, West Indies

Spondylolysis

Patient B

- Stress Fracture of the pars interarticularis of lumbar vertebrae
  - Can be unilateral or bilateral
  - +/- neurological symptoms
- Presents with back pain worsens with movement
- Causes by chronic overextension with rotation movements of the lumbar vertebrae
- On Oblique view of X-ray scotty dog has a collar



In simple spondylolysis, Scottie dog appears to be wearing a collar.

Scotty Dog with Collar


X-ray of Lumbosacral region (lateral view) showing scotty dog with colllar

Spondylolisthesis

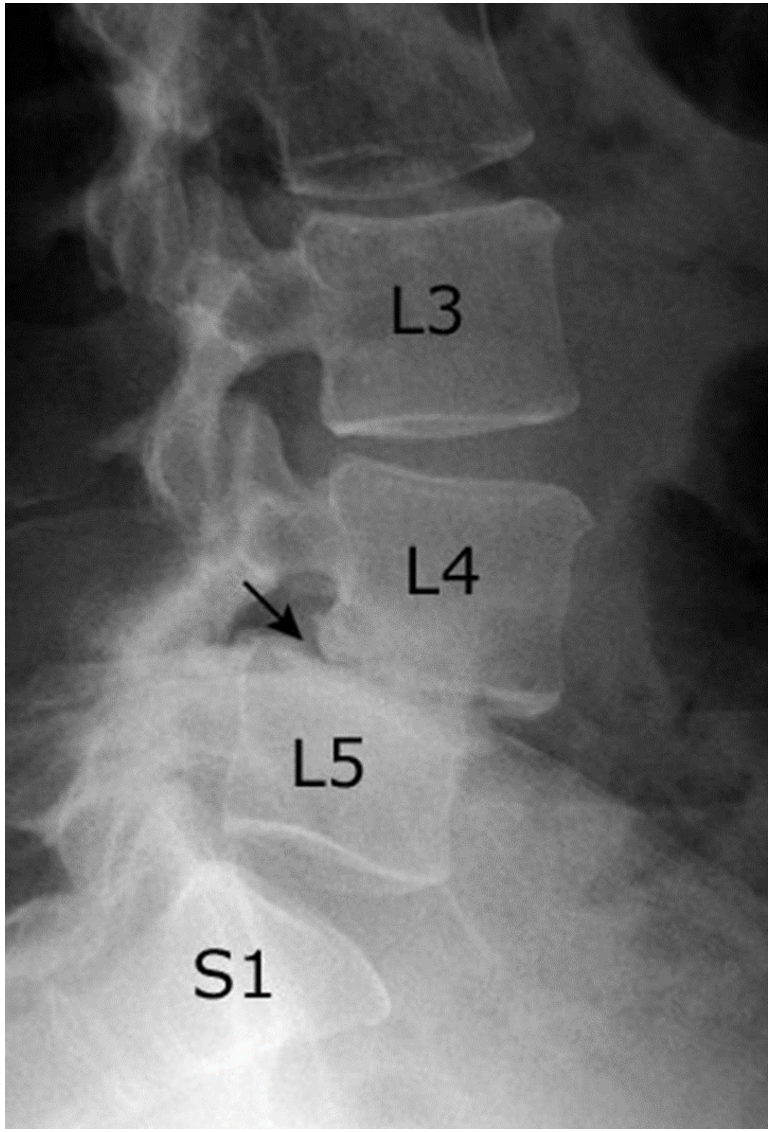
Patient B

- Anterior Stepping/slippage of vertebral body

- If occurring with spondylolysis (especially bilateral; fractures), then referred to as **spondylolysis with spondylolisthesis** or **spondylolytic spondylolisthesis**

- Slipping can result in narrowing of intervertebral foramina → spinal nerve impingements → neurological symptoms

In spondylolisthesis, Scotty dog appears decapitated



X-ray of Lumbosacral region (lateral view) showing anterior displacement of L4 on L5

Clinical Case 5

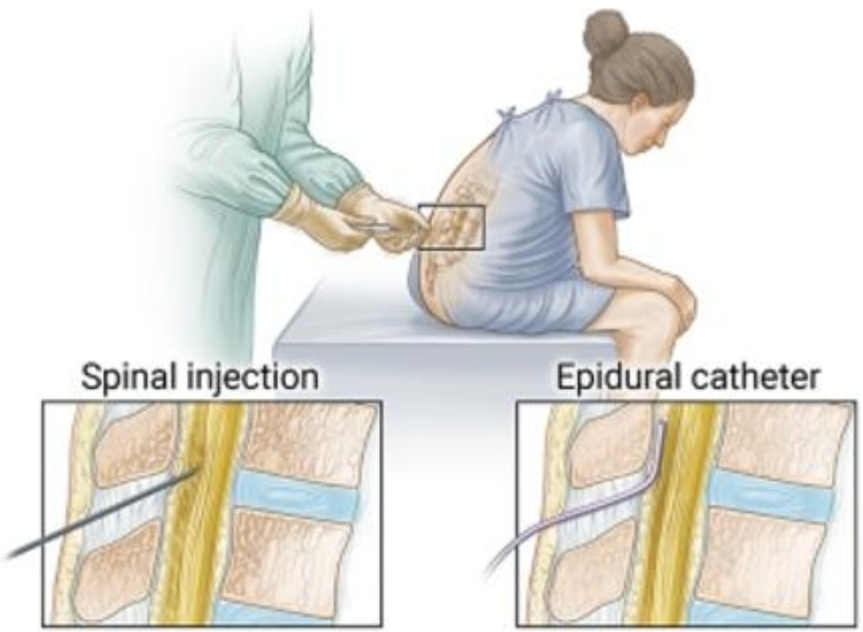
A primigravida (pregnant for the first time) woman comes to the emergency room because of labor pain. After further assessment she is subsequently admitted to the labor and deliver unit. Request for **pharmaceutical pain relief** as described in her birth plan.

8 hours later she delivers a healthy pair of twins via an uncomplicated vaginal delivery. Physical examination of one of **newborns shows a short-webbed neck.**.

Analgesia

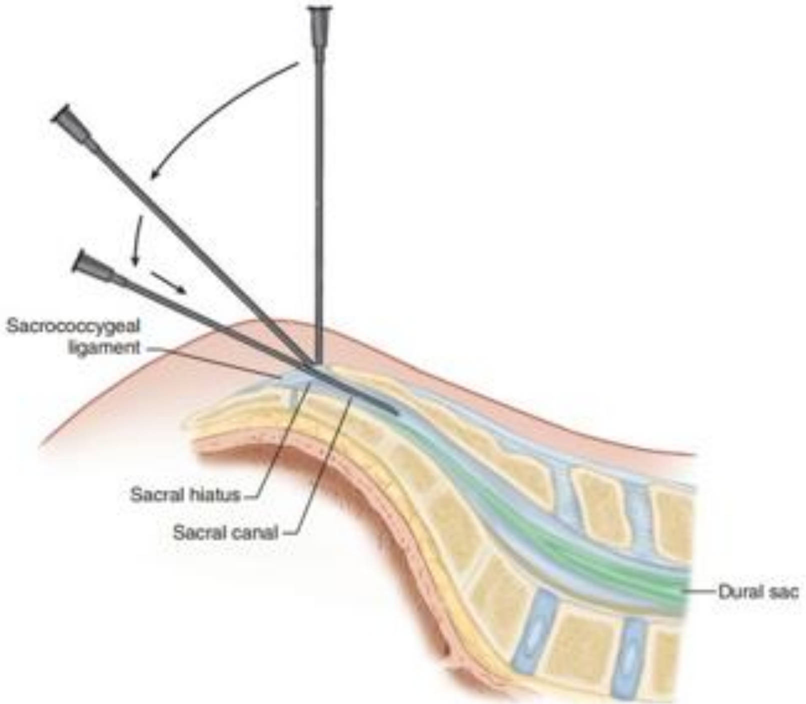
Epidural (Lumbar) Analgesia

- Same positioning as a lumbar puncture
    - Find Supracristal Line to count vertebrae
- “One Pop” felt. Which ligament?
- Nerves are within epidural space and can be anesthetized



Sacral Analgesia (Caudal Anesthesia)

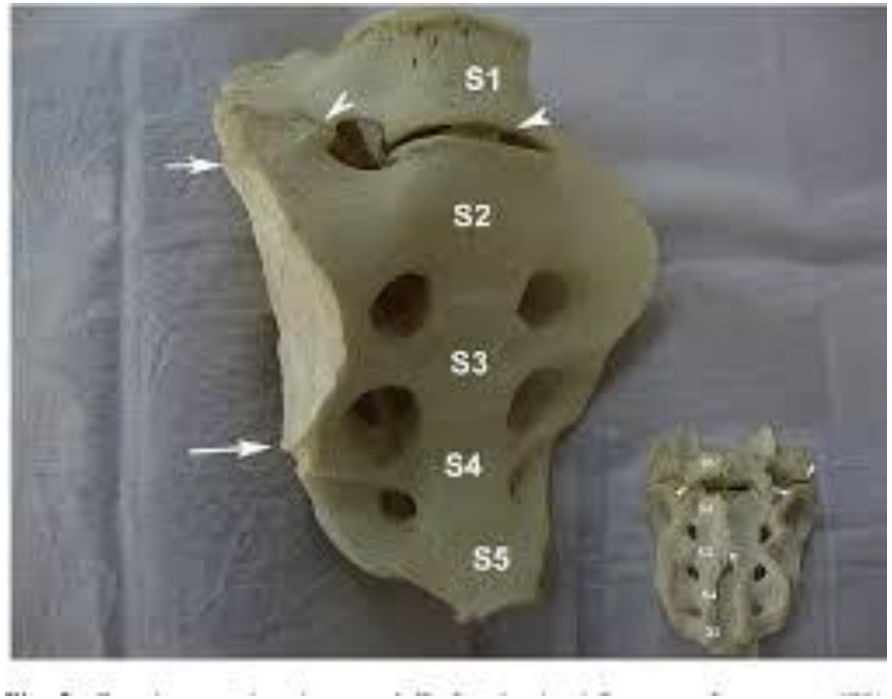
- Provides anesthesia to the caudal nerve roots
- Needle inserted into sacral hiatus
- Can be done under ultrasound guidance



Abnormal Vertebrae Count?

- Congenital Defect is possible making it difficult to assess vertebrae
- Sacralization of Vertebrae → partial fusion of the LV vertebrae to sacrum
- Lumbarization of vertebrae → partial separation of the S1 vertebrae from the sacrum

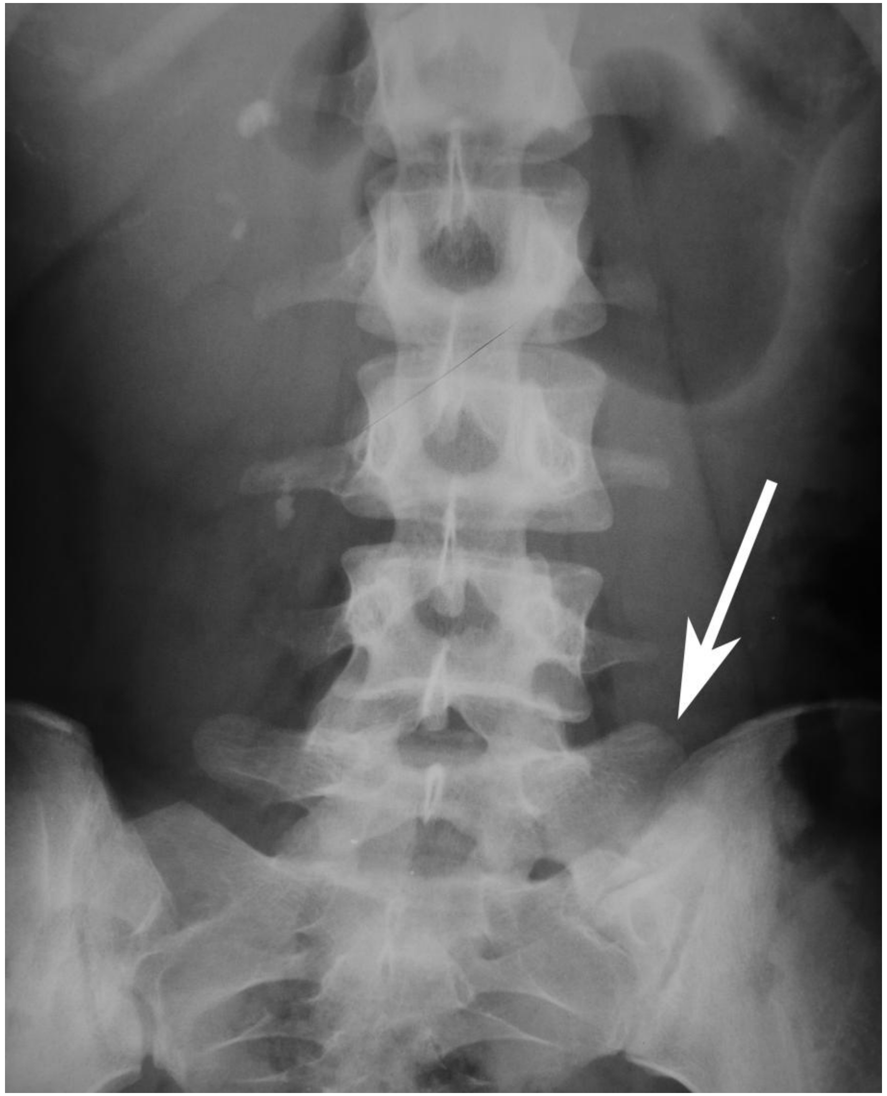
Sacralization



Lumbarization

St. George's University
Grenada, West Indies

Abnormal Vertebrae Count?



Frontal view X-ray of lumbosacral region displaying **Sacralization**

Frontal view X-ray of lumbosacral region displaying **Lumbarization**

St. George's University
Grenada, West Indies

Klippel Feil

- Congenital fusion of cervical vertebrae leading to a shortened neck
- Fusion can include other vertebrae
- Physical features
    - Shortened neck
    - decreased range of motion in the head and neck area
    - low hairline at the back of the head
- Associated with **scoliosis** and **hemivertebrae**

Co-twin Patient

Lateral view X-ray showing fusion of cervical vertebrae 4 -7

St. George's University
Grenada, West Indies

Hemivertebrae

- Half of the vertebra completely fails to form
- Caused by improper segmentation of sclerotomes
- Leads to the development of **Scoliosis (Congenital)**
    - Sharp angulation of the spine
    - Lateral deviation of vertebrae
    - Rotation of Vertebrae

Frontal view X-ray of vertebral column, arrows show hemivertebrae

Questions Email

mabdelra@sgu.edu or

Ask-anat@sgu.edu